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4/5/2026

ISE331-HW5

10.1. Compare and contrast routers and switches by describing the advantages and disadvantages of each.

Routers connect different networks and use IP addresses to forward data. Switches work inside the same local network and use MAC addresses. Switches are usually faster and cheaper, while routers are more powerful but more expensive. Routers and switches are both important network devices, but they have different purposes. A switch is mainly used in a local area network to connect devices and send data based on MAC addresses, so it is fast and efficient for communication within the same subnet. It is also cheaper and easier to use. However, it cannot connect different networks. A router, on the other hand, connects multiple networks and uses IP addresses to decide where to send data. It provides more control over traffic but is usually more expensive and slightly slower. In general, switches are better for local communication, while routers are needed for communication between networks.

11.1. A co-worker with a strong Windows background is having difficulty understanding Linux file permissions. Describe Linux file permissions to him. Compare Linux file permissions to Windows file permissions.

Linux permissions are divided into owner, group, and others. Each group has read, write, and execute permissions, shown as rwx rwx rwx. Linux permissions are simple and often managed by command line, while Windows permissions are more detailed and usually controlled through a graphical interface.

11.2. How are authentication and authorization alike and how are they different. What is the relationship, if any, between the two?

Authentication is used to verify a user's identity, while authorization is used to decide what the user can do. Both are used for access control, and authentication happens before authorization. Authentication and authorization are closely related but have different roles in security. Authentication is the process of verifying who the user is, such as logging in with a password or fingerprint. Authorization determines what actions the user is allowed to perform after they are authenticated. They are similar because both are used to control access to systems, but they are different in purpose. Authentication must happen first, because the system needs to know the

user's identity before giving permissions. Without authentication, authorization cannot work properly.

13.1. Discuss the differences between an anomaly-based and a misuse-based detection model. Which would you use to protect a corporate network of 10,000 users? Why would you choose that model?

An anomaly-based model detects unusual behavior and can find new attacks, but it is complex and produces many false positives. A misuse-based model detects known attack patterns, is easier to use, and has fewer false positives, but it cannot detect new attacks without known signatures.

An anomaly-based detection model focuses on identifying behavior that is different from normal patterns, which allows it to detect new or unknown attacks. However, it is more complex and often produces many false positives, which can be difficult to manage. A misuse-based model, on the other hand, looks for known patterns of attacks, making it easier to implement and more accurate for known threats, but it may miss new attacks without existing signatures. For a corporate network with 10,000 users, I would choose a misuse-based model because it is more stable and produces fewer false alarms, which is important in a large system. In practice, combining both models would provide better security.

15.1. Compare and contrast port scanning and ping sweeps.

Ping sweeps are used to find active devices in a network, while port scanning is used to find open ports and running services. Both are used for network information gathering.

Port scanning and ping sweeps are both techniques used to collect information about a network, but they focus on different aspects. A ping sweep is used to identify which IP addresses are active, meaning which devices are online. Port scanning is used to check which ports are open on a device and what services are running. The similarity is that both are used for network discovery, but the difference is that ping sweeps identify active hosts, while port scanning provides more detailed information about services. Together, they are often used in the early stage of network analysis or attacks.

15.2. What is the best practice to employ to mitigate malware effects on a machine?

To reduce malware effects, it is important to use antivirus software and keep it updated, and also keep the operating system and applications updated.

The best way to mitigate malware effects on a machine is to maintain good security practices. First, antivirus software should be installed and regularly updated so that it can detect and remove malware. Second, the operating system and all applications should be kept up to date,

because many malware attacks take advantage of known vulnerabilities in outdated software. By combining updated security software and regular system updates, the risk and impact of malware can be greatly reduced.